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SECURITY CLASSIFICATION OF THIS PAGE *(When Data Entered)*

REPORT DOCUMENTATION PAGE		READ INSTRUCTIONS BEFORE COMPLETING FORM
1. REPORT NUMBER TOP 2-2-501	2. GOVT ACCESSION NO. A092393	3. RECIPIENT'S CATALOG NUMBER
4. TITLE *(and Subtitle)* US ARMY TEST AND EVALUATION COMMAND TEST OPERATIONS PROCEDURE "SWIMMING TESTS OF WHEELED AND TRACKED VEHICLES"		5. TYPE OF REPORT & PERIOD COVERED Final 6. PERFORMING ORG. REPORT NUMBER
7. AUTHOR*(s)*		8. CONTRACT OR GRANT NUMBER*(s)*
9. PERFORMING ORGANIZATION NAME AND ADDRESS US ARMY ABERDEEN PROVING GROUND (STEAP-MT-M) ABERDEEN PROVING GROUND, MARYLAND 21005		10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS DARCOM-R 310-6
11. CONTROLLING OFFICE NAME AND ADDRESS US ARMY TEST AND EVALUATION COMMAND (DRSTE-AD-M) ABERDEEN PROVING GROUND, MARYLAND 21005		12. REPORT DATE 18 November 1980 13. NUMBER OF PAGES 15
14. MONITORING AGENCY NAME & ADDRESS*(if different from Controlling Office)*		15. SECURITY CLASS. *(of this report)* Unclassified 15a. DECLASSIFICATION/DOWNGRADING SCHEDULE

16. DISTRIBUTION STATEMENT *(of this Report)*

Approved for public release; distribution unlimited.

17. DISTRIBUTION STATEMENT *(of the abstract entered in Block 20, if different from Report)*

18. SUPPLEMENTARY NOTES

19. KEY WORDS *(Continue on reverse side if necessary and identify by block number)*

Amphibious vehicles
Mobility
Tactical vehicles
Tracked vehicles
Vehicles
Water facilities
Wheeled vehicles

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

Provides detailed tests to assess the swimming abilities of tactical wheeled and tracked vehicles.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-101
*Test Operations Procedure 2-2-501 November 18, 1980
AD No. A092393

SWIMMING TESTS OF WHEELED AND TRACKED VEHICLES

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1. SCOPE. This TOP describes tests to assess the performance of tactical wheeled and tracked vehicles in water, including personnel carriers, cargo carriers, trucks, trailers, etc., that have either built-in floating/swimming capability or kits to enable them to float or swim.

Land performance of such vehicles is evaluated in accordance with pertinent standard TOPs in Volume 2, while fording tests are described in TOP 2-2-612.[1]** Tests of amphibious vehicles, boats, barges, and motorcraft are described in TOP 9-2-251[2].

*This TOP supersedes TOP/MTP 2-2-501, 7 September 1967.

**Footnote numbers match reference numbers in Appendix D.

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2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

ITEM	REQUIREMENT
Water facilities: various conditions and sites (depending on item and test phase); ranges from still water with current, rough water, etc.	Refer to Appendix A
Safety equipment, as applicable (life rings and jackets, emergency signaling device, etc.)	
Marker buoys	
Towlines	
Sighting poles	
Safety patrol boat(s)	
Cameras (movie and still) and film	

2.2 Instrumentation.

ITEM	MAXIMUM PERMISSIBLE ERROR OF MEASUREMENT*
Device to measure vehicle speed (time/distance)	±0.2 km/hr
Engine tachometer	±0.5% full-scale reading
Timing device	±0.1 second
Device to measure turning radius	±1 meter
Current speed indicator	±0.3 knot
Device to measure freeboard	±2.5 cm
Device to measure pitch and roll	±0.5 degree

*Values may be assumed to represent ± 2 standard deviations; thus, the stated tolerances should not be exceeded in more than 1 measurement of 20.

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ITEM	MAXIMUM PERMISSIBLE ERROR OF MEASUREMENT
Devices to measure wind speed and direction and air temperature (meteorological data)	±1.6 km/hr (wind speed) ±1° (wind direction) ±1° C (air temperature)

3. PREPARATION FOR TESTS. Inspect the test item as described in applicable sections of TOP 2-2-505[3], and determine its physical characteristics as described in TOP/MTP 2-2-500[4]. Record the following:

a. Nomenclature
b. Manufacturer
c. Serial number
d. Model number
e. Hull material
f. Method of water maneuvering (as applicable):

(1) Steering (rudder, tracks)
(2) Propulsion:

Propeller, when applicable:
Number of propellers
Number of blades
Diameter and pitch

Tracks (standard or with modification kits that include water-propulsion cups, etc.)

Wheels

Water jet

4. TEST CONTROLS.

4.1 Safety. Observe the following:

a. Review all instructional material (including system support packages) issued with the test vehicle, and review reports of previous tests on vehicles of the same type.

b. Review the safety statement provided by the developer to determine whether any hazards have been identified. If hazards do exist, write the test plan to include subtests suitable for evaluating them.

c. Make sure applicable Standing Operating Procedures (SOPs) and Test Operations Procedures (TOPs) are available at the test site.

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d. Make sure all personnel involved in testing are thoroughly familiar with the provisions of pertinent SOPs and are fully capable of implementing them before any testing in water is begun.

e. Use approved life jackets and life-saving equipment. Make sure the driver/operator understands the behavior of vessels in water, as well as the effects of wind and current. Make sure the engine is in good condition; if possible, do not permit it to stall.

Safety procedures shall include, but shall not be limited to, the following conditions:

(1) Purpose of safety procedure (protection of life, property, etc.)

(2) Procedure before entering water

(3) Procedure during entry into water

(4) Procedure while afloat

(5) Procedure during exit from water

(6) Procedure after exit from water

(7) Procedure in emergencies while afloat (stalled engine, person overboard, bilge pump failure, swamping of vehicle, unpredicted/unforeseen circumstances)

(8) Placement of load in and covering of cargo compartment

(9) Procedure for determining unsafe loading conditions

(10) Personnel authorized aboard during test

(11) Number of personnel in cab (generally limited to design capacity of cab)

(12) Provision for and use of salvage and rescue equipment

(13) Provision for and use of water safety equipment (life jackets, rings, and preservers, emergency bilge pumps, marker buoys, signal equipment, emergency signals, etc.)

(14) Assignment of general and specific responsibilities, and delegation of appropriate authority to each individual directly involved with or in support of water tests.

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4.2 Water and Wind Restrictions. For certain phases of the marine engineering tests, such as speed, towline pull, and turning radii, use bodies of water that have negligible current velocities, and if available, sufficient depth and width to prevent adverse hydrodynamic effects. (Calculation of these effects on vehicular performance is complex; investigations of various sources reveal, however, that water depth 6 times the vehicle draft[5], or in another case[6], about 12 meters for 2-1/2-ton amphibious trucks, is preferred for accurate speed tests. Unavailability of suitable depths may compel compromise.) Neither speed trials nor turning radii can be obtained accurately in currents. Furthermore, wind velocities that exceed 15 km/hr are a serious handicap to obtaining accurate data. Hence, still water of suitable depth is imperative for comparative speed tests, and wind must be minimal during trial runs.

5. PERFORMANCE TESTS. Conduct preliminary operations on land as described in the applicable sections of TOP 2-2-505. Meet the following conditions and restrictions before performing water tests:

a. Make sure a safety patrol boat with qualified rescue personnel and equipment are available.

b. Establish communications to assure an immediate response of emergency assistance.

c. Attach a heavy duty cable bridle to each end of the test vehicle, and tie a loop to the top side of the vehicle for quick release (the bridle is then available for recovery operations).

d. Train and prepare vehicle operators to use underwater breathing apparatus in case of an emergency.

e. If testing in water deep enough to completely inundate the vehicle, attach marker buoy(s) to the vehicle to locate it and any test crew/driver that are trapped.

5.1 Preliminary Water Operations.

5.1.1 Method. Subject the test vehicle to water tests as follows:

a. Operate in still water to insure proper functioning of controls and components, particularly with respect to adequate clearance for the driving members, operation of bilge pumps, and tightness of hull.

b. Perform tests when there is little or no wave activity, current flow, or wind.

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c. Operate empty test vehicle in the water for at least 2 hours. Stop and start the engine(s) at least three times during this period. Determine the maximum safe forward speed and minimum turning radius.

d. Take and retain motion pictures of all operations.

5.1.2 Data Required. Record the following:

a. Vehicle weight
b. Water depth
c. Amount of freeboard (at four corners)
d. Leakage rate, if any
e. Environmental conditions:

(1) Wind speed and direction
(2) Air temperature
(3) Current flow and direction
(4) Estimated wave height, if any

f. Trim and list
g. Performance of controls
h. Operator comments and interview (on applicable forms)
i. Vehicular malfunctions, if any
j. Maximum safe launching and landing angles of test vehicle
k. Maximum safe forward speed
l. Minimum turning radius

5.2 Load Distribution and Effects of Unequal Load.

5.2.1 Method. Load the test vehicle with its rated payload, making sure that the payload is adequately distributed and secured. The payload shall also be appropriate in weight and type for the test item and operating conditions. Repeat the steps listed in 5.1 with the test vehicle loaded. Repeat the sequence again with a rated payload of personnel or cargo, as applicable, incrementally shifted to both sides and/or forward and rearward of the test vehicle to determine limits of load distribution.

5.2.2 Data Required. Record the following for each applicable test condition:

a. Water depth
b. Environmental conditions:

(1) Wind velocity and direction
(2) Current flow and direction
(3) Estimated wave height, if any

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c. Performance of controls
d. Vehicular malfunctions, if any
e. Leakage, if any
f. Amount of freeboard
g. Trim and list
h. Maximum safe launching/landing angles of test vehicle
i. Maximum safe forward speed
j. Minimum turning radius
k. Type and weight of load
l. Swimming directional control
m. Loss of buoyancy
n. Spillage of fuel, etc.
o. Crew egress problems, if any

5.3 Maximum Water Speed Test.

5.3.1 Method. Perform the speed test as follows:

a. Prepare the test course in a facility similar to that described in Appendix A.

b. Instrument the test vehicle with a tachometer, as applicable.

c. Conduct the speed test over a course for a least 150 meters. The course shall be as close to shore as the depth limitations permit. Use state-of-the-art instrumentation to directly measure vehicle speed, or set up dual sighting poles at each end of the measured course, using surveying techniques. Place buoys in the water outside the sighting lines to mark a course at right angles to the sighting poles. Do not use buoys for timing distances, but only for assisting the operator in following the shortest course during timed runs.

d. Operate the test vehicle without a payload over the test course for at least three runs in each direction at full throttle for each transmission gear.

e. Repeat steps c and d with rated payload. Record the type and weight of the load.

f. Take and retain motion pictures of all operations.

5.3.2 Data Required. Record the following:

a. Water depth
b. Environmental conditions:

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(1) Wind speed and direction
(2) Current flow and direction
(3) Estimated wave height, if any

c. Distance of course
d. For each run, direction, and gear position:

(1) Number of run (1, 2, etc.)
(2) Direction
(3) Gear (1st, 2nd, etc.)
(4) Speed of engine
(5) Time required to traverse course
(6) Water speed attained
(7) Noted hazards
(8) Amount of freeboard

e. Type and weight of load

5.4 Towline Pull Test.

5.4.1 Method. Determine the towline pull of the test vehicle as described in applicable sections of TOP 2-2-604[7], with the empty test vehicle pulling against a fixed object on shore (or anchored) to develop pull versus engine speed curve(s) in pre-determined operating gear range(s).

Repeat the above sequence with rated payload. Record type and weight of load.

5.4.2 Data Required. Collect and record towline pull data as described in applicable sections of TOP 2-2-604.

5.5 Vehicular Direction Control Tests. Use a facility similar to that described in Appendix A for these tests.

5.5.1 Control While Turning. To test control while turning, select or establish reference points so that the turning radius can be obtained with sufficient accuracy by measuring distances and angles between the reference points and the vehicle while it turns; or use other state-of-the-art instrumentation for measuring vehicle turning radii.

5.5.1.1 Method.

a. If applicable, instrument the test vehicle with a tachometer.

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b. Operate empty test vehicle over the course at maximum safe speed and maximum steer input, right and left, in a full circle. Repeat this procedure a sufficient number of turns in each direction to assure representative results.

c. Repeat step b with rated payload.

5.5.1.2 Data Required. Record the following:

a. Water depth
b. Environmental conditions:

(1) Wind speed and direction
(2) Current flow and direction
(3) Estimated wave height, if any

c. Vehicle and engine speed
d. Any hazards noted
e. Amount of freeboard
f. Type and weight of load
g. Distances and angles between the reference points and turning vehicle on graph paper
h. All difficulties encountered

5.5.2 Control in a Straight Corridor. One measure of controllability is the "corridor" within which an amphibian can be held on a straight course. Minimum corridor should therefore be measured over a distance sufficient to produce stabilized speed and steering effects.

5.5.2.1 Method. Establish a straight operating corridor of minimum distance sufficient to produce stabilized speed and steering effects for the test vehicle.

a. If applicable, instrument the test vehicle with a tachometer.

b. Operate empty test vehicle through the corridor directed into the current at full engine speed.

c. Repeat step b with rated payload.

d. Repeat steps b and c, operating with the current.

5.5.2.2 Data Required. Record the following:

a. Items required under paragraph 5.5.1.2 of this TOP.

b. Minimum corridor width

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5.6 Cooling Tests. NOTE: This subtest is not essential to every test program, but may be conducted at the discretion of the evaluator.

5.6.1 Method. Conduct these tests at maximum vehicle water speed and as described in applicable sections of TOP/MTP 2-2-607[8].

5.6.2 Data Required. Record the following:

a. Adequacy of cooling system
b. Insulation adequacy for hot spots
c. Stabilized temperatures of:

(1) Engine coolant and oil temperature
(2) Transmission oil
(3) Air intake
(4) Exhaust
(5) Crew compartments

d. Vehicle and engine speed

5.7 Fuel Consumption. NOTE: This subtest is not essential to every test program, but may be conducted at the discretion of the evaluator.

5.7.1 Method.

a. On water, obtain fuel-consumption data for empty test vehicle as described in applicable sections of TOP 2-2-603[9], in the gear combination that yields maximum water speed, at discrete points in the operating speed range of the engine.

b. Repeat with rated payload.

5.7.2 Data Required. Record data as described in TOP 2-2-603.

5.8 Ventilation Evaluation.

5.8.1 Method. On water, evaluate the ventilation of each crew compartment and cargo space by measuring toxic fumes concentration, as described in applicable sections of TOP 2-2-614[10], under several wind conditions and vehicle speeds.

5.8.2 Data Required. Record data as required in TOP 2-2-614.

5.9 Bilge Pump Evaluation.

5.9.1 Method. Evaluate bilge pumps during testing by inspecting them periodically to ensure proper and adequate operation. Place excess water in the bilges and measure the quantity pumped out for a known period of time from the pump outlet(s).

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5.9.2 Data Required. Determine and record the following:

a. Ability of bilge pumps to prime themselves and operate effectively under:

(1) Shifting of full loads to extreme positions
(2) Other conditions as deemed appropriate

b. Overall performance of bilge pumps (flow rate)
c. Accessibility of bilge pump controls
d. Freedom of bilge pump linkage from interference
e. Adequacy of screening of bilge pump intakes

5.10 Winch Tests.

5.10.1 Method. Conduct winch tests as described in applicable sections of TOP 2-2-712[11], and determine ability of winch to assist vehicle ashore when immobilized.

5.10.2 Data Required. Collect and record data as described in applicable sections of TOP 2-2-712.

5.11 River Crossing Tests. NOTE: An appropriate site should include various soil conditions and embankment grades consistent with the test vehicle design.

5.11.1 Method.

a. Establish a river-crossing site with maximum current velocities as described by test vehicle specifications.

b. When appropriate to the design, attempt stream crossing, aided by runners on a fixed cable; this method may facilitate crossing that would otherwise be impossible.

c. When vehicle specifications so require, operate empty test vehicle across the river at least five times, and determine its ability to maneuver under full control in a current of stated velocity.

d. When appropriate to the design of the test vehicle, connect empty test vehicle to the fixed cable, and operate it across the river at least five times. Determine stability limitations of the test vehicle. NOTE: Because of their design, some vehicles have a tendency to capsize by action of the current when the upper part of the vehicle is restrained by a crossing cable.

e. Repeat steps c and d with rated payload. Record the type and weight of the load.

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5.11.2 Data Required. Record the following data for each performance, as described in 5.11.1 above:

a. Description of crossing site:

(1) Type of approach(es)
(2) Angles of approach(es)
(3) Water depth
(4) Current speed

b. Ability of test vehicle to maneuver under full control
c. Vehicular stability limitations
d. Type and weight of load
e. All difficulties encountered

6. DATA REDUCTION AND PRESENTATION.

6.1 Summarize and evaluate data obtained for each testing performance.

6.2 Submit appropriate graphs summarizing test data.

6.3 Identify all photography (movie and still).

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APPENDIX A
WATER FACILITIES

At Aberdeen Proving Ground, the Spesutie Narrows, with a channel 3 to 4 meters deep, is used for shallow water tests.

The moderately sloped amphibian ramp (see Figure A-1) extends into the water sufficiently to permit safe launching of test vehicles that have unknown flotation characteristics. For entrance and exit tests, slopes of the interface area between the water and land are constructed by grading.

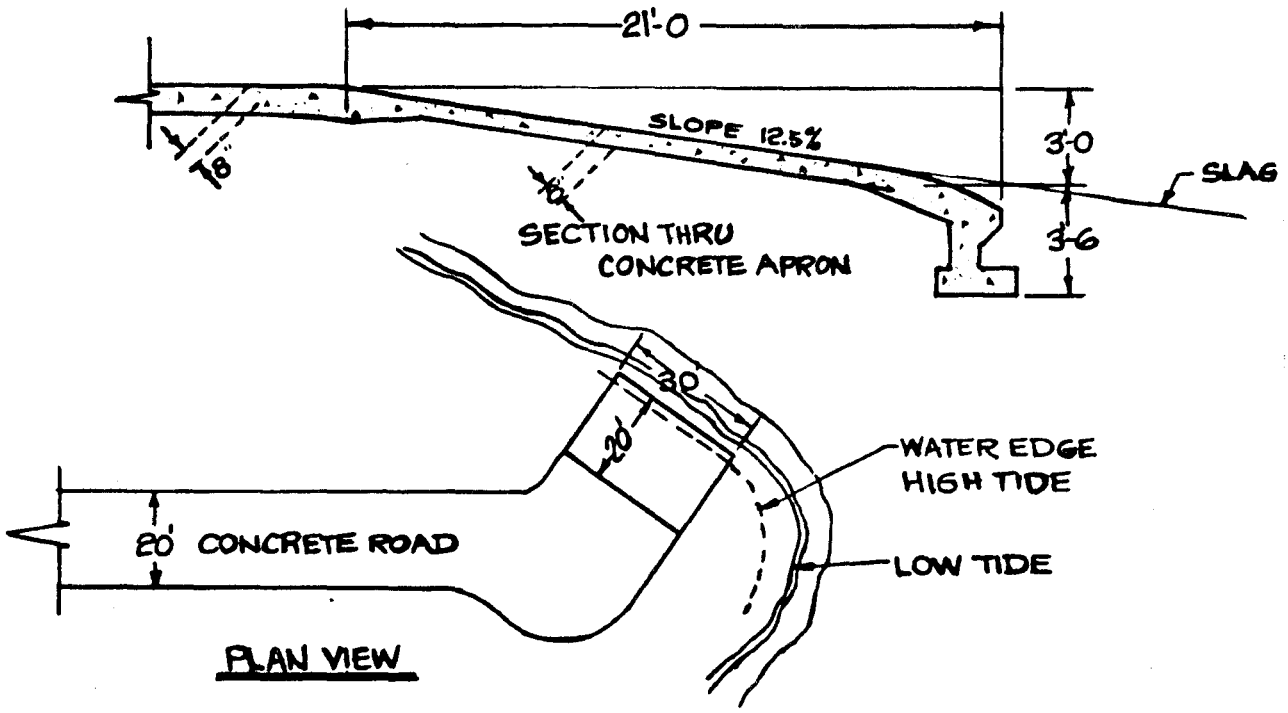


FIGURE A-1. PLAN VIEW AND SECTION, AMPHIBIAN RAMP, ABERDEEN PROVING GROUND.

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APPENDIX B
MAINTENANCE REQUIREMENTS

During testing, determine the maintainability of each test item, collect and record data as described in applicable sections of TECOM Supplement 1 to DARCOM 700-15.[12] Analyze, evaluate, and record special maintenance requirements arising from salt water and atmospheric conditions.

Measure water contamination of fuel and lubricants.

Evaluation of Corrosion.

Although a vehicle may be designed for inland or protected waters, it must be capable of withstanding the corrosive effects of salt water and atmosphere, because many so-called inland waters are actually estuaries, straits, or fiords. Electrolytic effects are especially important because of the rapid deterioration of metallic parts when in proximity to dissimilar metals. The effect is accelerated by temperature, and is greatest when the two metals are widely separated on the electromotive scales, e.g., magnesium and iron. Corrosion is further discussed in TOP 2-2-612.

If testing is conducted in salt or brackish water, periodically inspect the test vehicle visually for effects of corrosion. After testing is completed, clean the test item(s) thoroughly, including flushing with fresh water, and thoroughly examine it for effects of corrosion. NOTE: Completely tear down any components of the vehicle or related item that show evidence of corrosion, and inspect to determine the extent of damage from corrosive action. Take and retain still photographs of adverse findings, including extent of damage and all evidence.

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APPENDIX C
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